

SOC-5811 Week 6: Simulation

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REVIEW

- ▶ Sampling distributions



REVIEW

- ▶ Sampling distributions
- ▶ Statistical inference and p-values



REVIEW

- ▶ **Standard deviation** = a measure of the spread of data around the sample mean.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

- ▶ **Standard error** = a measure of the spread of sample means around the population mean; the standard deviation of the sampling distribution.

$$\text{Standard error} = \frac{\sigma}{\sqrt{n}}$$

SIMULATION

- ▶ Last week was **statistical inference**: We *observe* sample data → we estimate the most likely *unknown* population model that produced that sample data.



SIMULATION

- ▶ Last week was **statistical inference**: We *observe* sample data → we estimate the most likely *unknown* population model that produced that sample data.
- ▶ This week is **simulation**: We *know* the population model → we simulate all the sample data is could plausibly produce (i.e., we simulate the *sampling distribution*).



SIMULATION

Why is simulation an important concept/skill?

- ▶ Develop statistical intuition without needing math.



SIMULATION

Why is simulation an important concept/skill?

- ▶ Develop statistical intuition without needing math.
- ▶ We can use simulation to approximate the sampling distribution of data and propagate this to the sampling distribution of statistical estimates (aka **bootstrapping**).



SIMULATION

Why is simulation an important concept/skill?

- ▶ Develop statistical intuition without needing math.
- ▶ We can use simulation to approximate the sampling distribution of data and propagate this to the sampling distribution of statistical estimates (aka **bootstrapping**).
- ▶ Very important principle for more complex models: you can always simulate.



SIMULATION

- ▶ Remember: this is all about understanding how we do **statistical inference**.



SIMULATION

- ▶ Remember: this is all about understanding how we do **statistical inference**.
- ▶ What can I learn about the population model from the data I observe?



SIMULATION

- ▶ Remember: this is all about understanding how we do **statistical inference**.
- ▶ What can I learn about the population model from the data I observe?
- ▶ Are my sample parameters real, or could they plausibly have been observed by chance?



SIMULATION

Table 2. Children with Same-Sex Parents and Standardized Test Scores at the End of Primary Education

	All Children		Children Raised by Same-Sex Parents from Birth ^a		
	(1)	(2)	(3)	(4)	(5)
Child has same-sex parents (1 is yes)	.106*** (.019)	.054** (.018)	.104*** (.024)	.139*** (.023)	.147*** (.041)
Sex (1 is male)	-.004* (.002)	-.009*** (.002)	-.003 (.002)	-.007*** (.002)	
Ethnicity (ref: both parents born in NL)					
One parent NL, other Western country		.004 (.003)		.004 (.003)	
One parent NL, other non-Western country		-.031*** (.005)		-.031*** (.005)	
Both parents not NL		-.111*** (.006)		-.111*** (.006)	
Parental education (1 is diploma SE)		.531*** (.004)		.532*** (.004)	
Log household income		.083*** (.002)		.084*** (.002)	
Mean parental age		.018*** (.000)		.018*** (.000)	
Family size		-.066*** (.001)		-.066*** (.001)	
Family structure (ref: married parents)					
Cohabiting parents		-.073*** (.002)		-.073*** (.002)	
Other		-.367*** (.012)		-.154*** (.032)	
Fixed effects					
Birth year	Yes	Yes	Yes	Yes	
Neighborhood	No	Yes	No	Yes	
Method	OLS	OLS	OLS	OLS	CEM ^b
Number of children	1,204,692	1,204,692	1,195,624	1,195,624	48,388
Number of children with same-sex parents	2,971	2,971	1,390	1,390	1,390
R ²	.004	.108	.004	.109	.121

SIMULATION

Table 1

Estimates of the Effect of Veteran Status on Homeownership by Race (1960)

	Outcome	All	White	Black
Stage I	Veteran status	.396*** (.005)	.409*** (.005)	.274*** (.009)
Reduced Form	Homeownership	.048*** (.003)	.051*** (.003)	.020** (.005)
IV	Homeownership	.121*** (.008)	.125*** (.009)	.071*** (.019)
OLS	Homeownership	.114*** (.006)	.104*** (.007)	.080*** (.011)

THIS WEEK: R SKILLS

► Functions in R



THIS WEEK: R SKILLS

- ▶ Functions in R
- ▶ Simulating in R



THIS WEEK: R SKILLS

- ▶ Functions in R
- ▶ Simulating in R
- ▶ Bootstrapping in R



THIS WEEK: R SKILLS

- ▶ Functions in R
- ▶ Simulating in R
- ▶ Bootstrapping in R
- ▶ Introduction to multivariable models



FUNCTIONS IN R

```
log(5)
```

```
## [1] 1.609438
```

FUNCTIONS IN R

```
new_function <- function(x) {  
  y <- x*10  
  return(y)  
}
```

FUNCTIONS IN R

```
new_function <- function(x) {  
  y <- x*10  
  return(y)  
}  
new_function(5)
```

```
## [1] 50
```

FUNCTIONS IN R

```
new_function <- function(x) {  
  y <- x*10  
  return(y)  
}  
sapply(c(5,10,15,20), new_function)
```

```
## [1] 50 100 150 200
```

SIMULATION

- ▶ Random variables



SIMULATION

- ▶ Random variables
- ▶ Distributions:



SIMULATION

- ▶ Random variables
- ▶ Distributions:
 - ▶ Normal distribution



SIMULATION

- ▶ Random variables
- ▶ Distributions:
 - ▶ Normal distribution
 - ▶ Binomial distribution

SIMULATION

- ▶ Random variables
- ▶ Distributions:
 - ▶ Normal distribution
 - ▶ Binomial distribution
- ▶ **Simulation of random variables in R.**

SIMULATION

X = the number of heads I get if I flip a coin 100 times.

X = Binomial($n=100$, $p=0.5$)

```
rbinom(n=1, size=100, prob=0.5)
```

```
## [1] 40
```

SIMULATION

```
rbinom(n=1, size=100, prob=0.5)
```

```
## [1] 49
```

```
rbinom(n=1, size=100, prob=0.5)
```

```
## [1] 55
```

```
rbinom(n=1, size=100, prob=0.5)
```

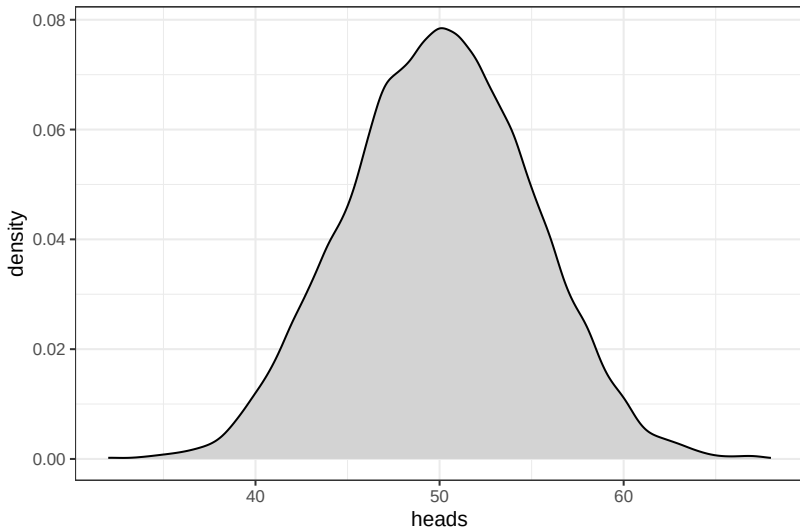
```
## [1] 47
```

SIMULATION

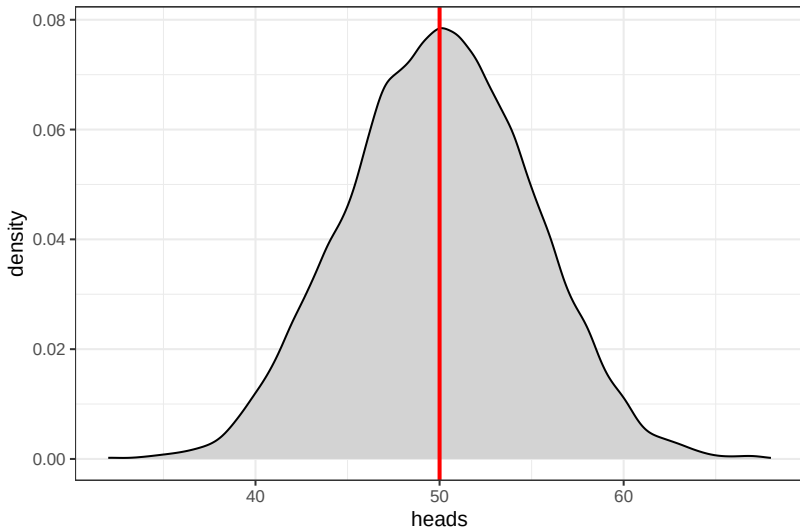
```
trials <- rbinom(n=10000, size=100, prob=0.5)
trials <- data.table(heads=trials)
head(trials)
```

```
##      heads
##      <int>
## 1:      47
## 2:      57
## 3:      43
## 4:      59
## 5:      47
## 6:      51
```

SIMULATION



SIMULATION



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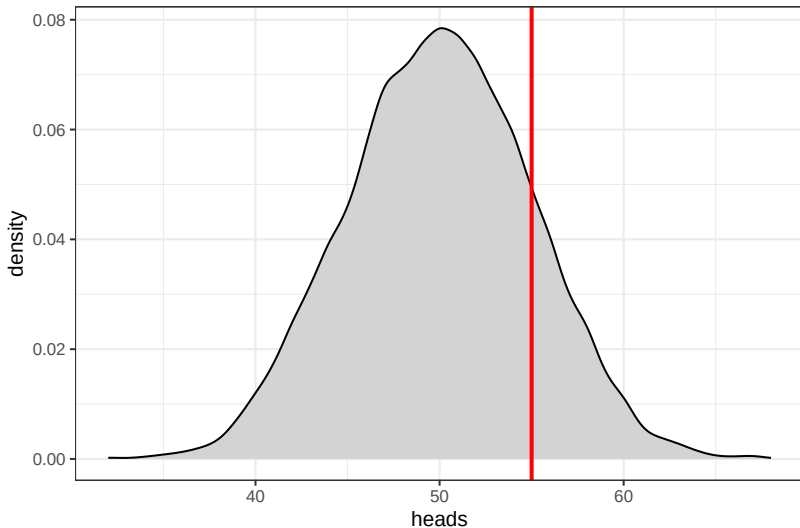
SIMULATION

Imagine flipping a coin 100 times and interpreting the results relative to some null probability model.

Null model: $P(\text{heads}) = 0.5$; this is a fair coin.

Results: 55 heads, 45 tails.

SIMULATION



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SIMULATION

```
pbinom(55, 100, 0.5, lower.tail = FALSE)
```

```
## [1] 0.1356265
```

SIMULATION

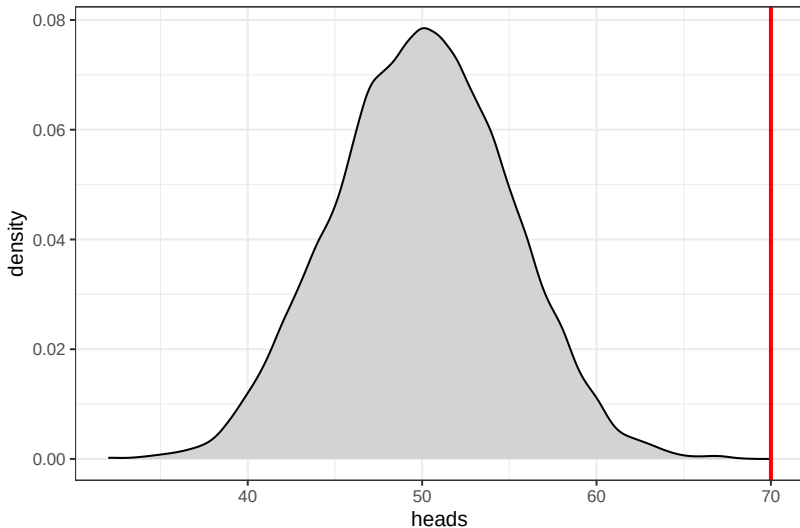
Imagine flipping a coin 100 times and interpreting the results relative to some null probability model.

Null model: $P(\text{heads}) = 0.5$; this is a fair coin.

Results: 55 heads, 45 tails.

Is the hypothesis that this is a fair coin consistent with our data? Could a fair coin plausibly produce our data?

SIMULATION



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SIMULATION

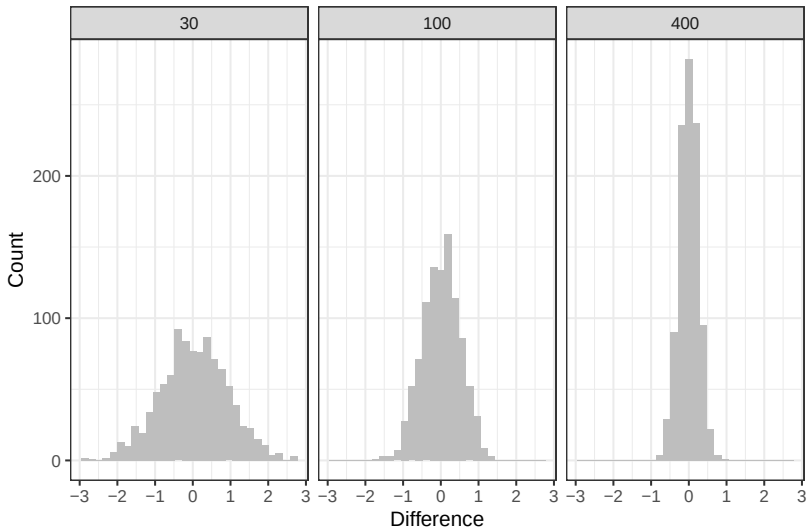
```
pbinom(70, 100, 0.5, lower.tail = FALSE)
```

```
## [1] 1.608001e-05
```

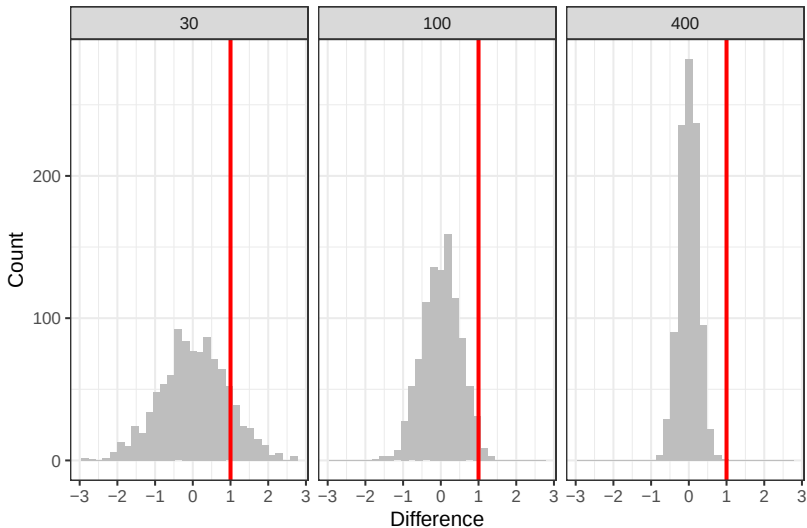
SIMULATION

```
mean_of_normal_trials <- function(i, ss) {  
  mean <- mean(rnorm(ss, 0, 5))  
  return(mean)  
}  
normal_trials <- function(ss) {  
  trials <- lapply(1:1000,  
                  mean_of_normal_trials,  
                  ss=ss)  
  trials <- data.table(sample_size=ss,  
                      mean=unlist(trials))  
  return(trials)  
}  
trials <- lapply(c(30,100,400), normal_trials)  
trials <- rbindlist(trials)
```

SIMULATION



SIMULATION



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BOOTSTRAPPING

- ▶ Simulating a sampling distribution when you **do not know** the underlying probability model.
- ▶ Randomly sample the **observed** data and calculate your statistic many times (“bootstrapping”).



BOOTSTRAP

```
psid <- fread('https://raw.githubusercontent.com/ngraetz/soc5811/refs/heads/main/data/psid.csv')  
psid <- psid %>%  
  filter(g3_house_value!=0)
```

BOOTSTRAP

```
mean_analytic <- psid %>%  
  summarise(mean=mean(g3_house_value)) %>%  
  pull(mean)  
se_analytic <- psid %>%  
  summarise(se=sd(g3_house_value)/sqrt(nrow(psid))) %>%  
  pull(se)  
mean_analytic
```

```
## [1] 388711.5
```

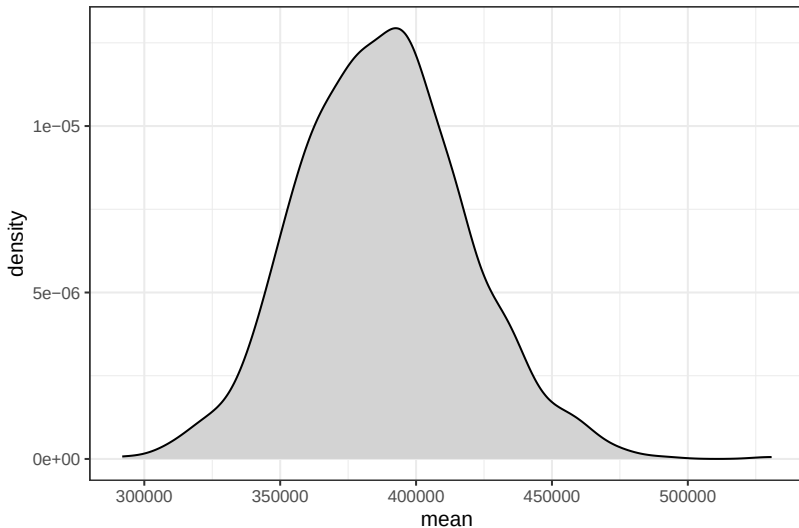
```
se_analytic
```

```
## [1] 31072.29
```

BOOTSTRAP

```
get_sample <- function(i) {  
  random_sample <- psid %>%  
    sample_n(size=nrow(psid), replace=T) %>%  
    summarise(mean=mean(g3_house_value))  
  return(random_sample)  
}  
samples <- rbindlist(lapply(1:draws, get_sample))
```

BOOTSTRAP



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BOOTSTRAP

```
## Calculate standard error  
se_bootstrap <- samples %>%  
  summarise(sd=sd(mean)) %>%  
  pull(sd)
```

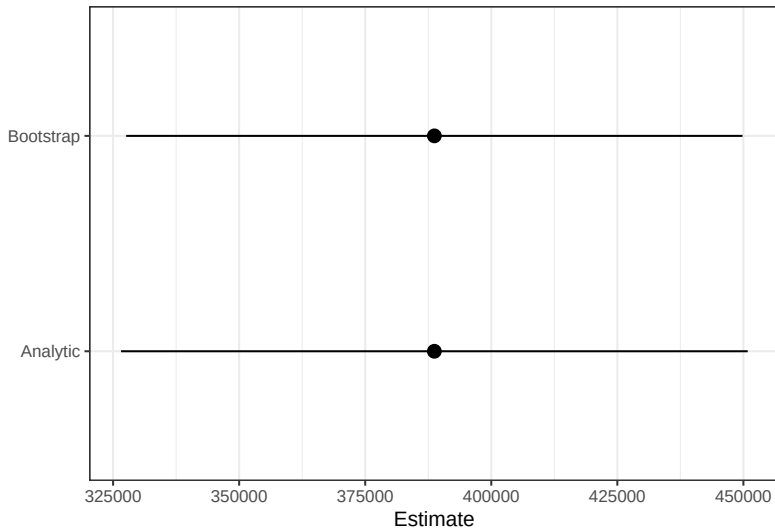
```
## Compare  
se_analytic
```

```
## [1] 31072.29
```

```
se_bootstrap
```

```
## [1] 30556.75
```

BOOTSTRAP



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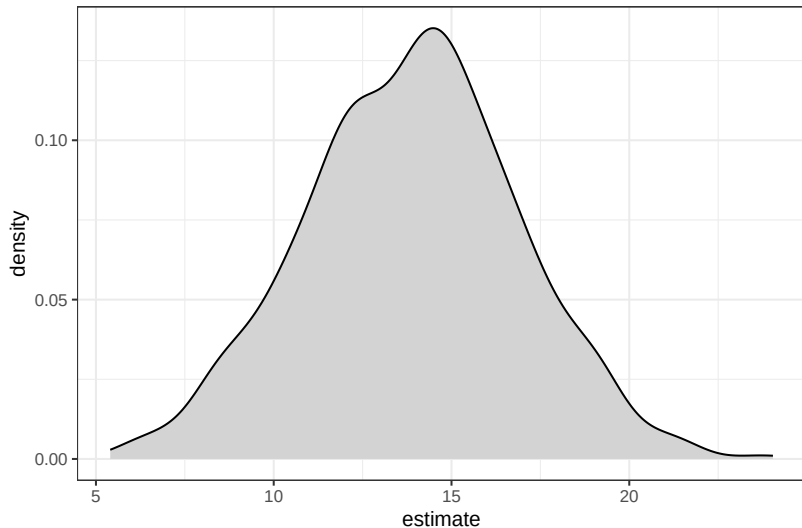
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BOOTSTRAP

```
psid_subset <- psid %>%  
  filter(!is.na(g1_edu_cat) & g3_house_value!=0) %>%  
  mutate(above_hs=ifelse(g1_edu_cat %in% c('less_hs', 'hs',  
                                           'less', 'more'))  
  
get_sample <- function(i) {  
  random_sample <- psid_subset %>%  
    sample_n(size=nrow(psid_subset), replace=T)  
  model <- lm(g3_house_value_percentile~above_hs,  
              data=random_sample)  
  out <- avg_slopes(model)  
  return(out)  
}  
coef_samples <- rbindlist(lapply(1:draws, get_sample))
```

BOOTSTRAP



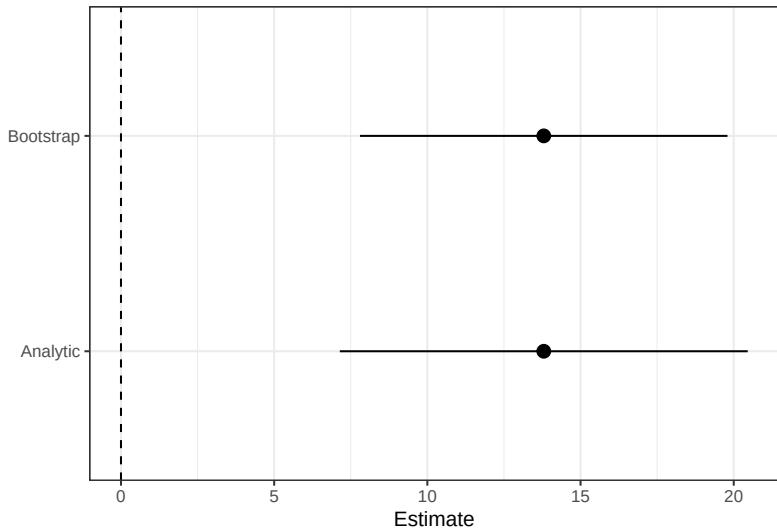
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BOOTSTRAP

```
## Compare to lm()  
model <- lm(g3_house_value_percentile~above_hs,  
            data=psid_subset)  
analytic_coefs <- as.data.table(avg_slopes(model))
```

BOOTSTRAP



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REVIEW

- ▶ Sampling distributions, statistical inference, and p-values.



REVIEW

- ▶ Sampling distributions, statistical inference, and p-values.
- ▶ What do we need to calculate?



REVIEW

- ▶ Sampling distributions, statistical inference, and p-values.
- ▶ What do we need to calculate?
 - ▶ **Standard errors / confidence intervals.**



REVIEW

How do we calculate standard errors / confidence intervals?

- ▶ **Analytic approach:** Use formulas.



How do we calculate standard errors / confidence intervals?

- ▶ **Analytic approach:** Use formulas.
- ▶ **Simulation approach:** Look at the variation in many random samples (with replacement) from the data, which can be taken as an approximation to the sampling distribution of our data collection process; that is, possible data we could have seen, had the data collection been conducted again in the same population.

REVIEW

```
bootstrap <- function(i) {  
  1. Randomly sample with replacement from  
    our data.  
  2. Calculate our target stat.  
}  
3. Bootstrap many times to approximate the  
   sampling distribution of our target stat.  
4. Standard error = standard deviation of  
   the sampling distribution.
```



REVIEW

$$y_i = \beta_0 + \beta_1 X$$

- ▶ Estimate standard errors / confidence intervals.



REVIEW

$$y_i = \beta_0 + \beta_1 X$$

- ▶ Estimate standard errors / confidence intervals.
- ▶ Estimate likelihood we may have seen a result as extreme or more extreme than our observed result if some null model were true.



REVIEW

$$y_i = \beta_0 + \beta_1 X$$

- ▶ Estimate standard errors / confidence intervals.
- ▶ Estimate likelihood we may have seen a result as extreme or more extreme than our observed result if some null model were true.
- ▶ Infer something about the population model!



REGRESSION WITH MULTIPLE VARIABLES

Why would we want to add multiple predictor variables?



REGRESSION WITH MULTIPLE VARIABLES

Why would we want to add multiple predictor variables?

- ▶ Prediction: few social science processes can be accounted for by a single explanatory variable (i.e., our predictions will be crude).



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- ▶ Comparison:



REGRESSION WITH MULTIPLE VARIABLES

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- ▶ Comparison:
 - ▶ Some descriptive comparisons involve more than one variable: “Within states, what is the average association between income and happiness?”



REGRESSION WITH MULTIPLE VARIABLES

Why would we want to add multiple predictor variables?

- ▶ Prediction: few social science processes can be accounted for by a single explanatory variable (i.e., our predictions will be crude).
- ▶ Comparison:
 - ▶ Some descriptive comparisons involve more than one variable: “Within states, what is the average association between income and happiness?”
 - ▶ Causal inference: omitted variable bias and confounding in interpreting a comparison as *the effect* of the independent variable on the dependent variable (i.e., *if I change the independent variable, then the dependent variable will change*).

SIMPSON'S PARADOX

Simpson's paradox

- ▶ Aggregation problem: an association may not be true for all subgroups (i.e., the problem is we are *generalizing* a high level pattern to lower levels).



SIMPSON'S PARADOX

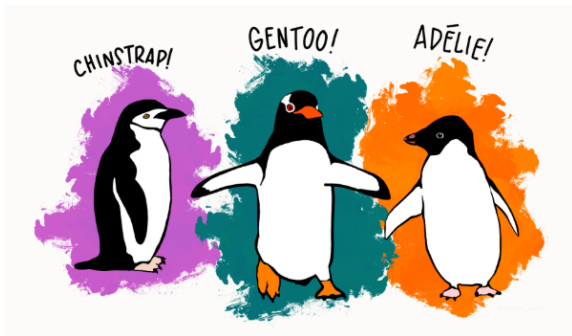
Simpson's paradox

- ▶ Aggregation problem: an association may not be true for all subgroups (i.e., the problem is we are *generalizing* a high level pattern to lower levels).
- ▶ Thinking about Simpson's paradox is useful for understanding why we need to be really specific about our research questions.

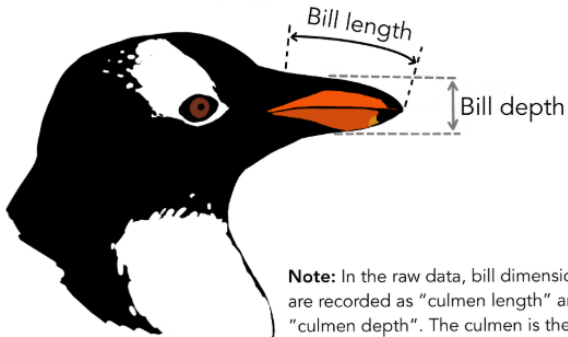


SIMPSON'S PARADOX

Data on 344 penguins were collected and made available by Dr. Kristen Gorman and the Palmer Station, Antarctica LTER, a member of the Long Term Ecological Research Network.



SIMPSON'S PARADOX



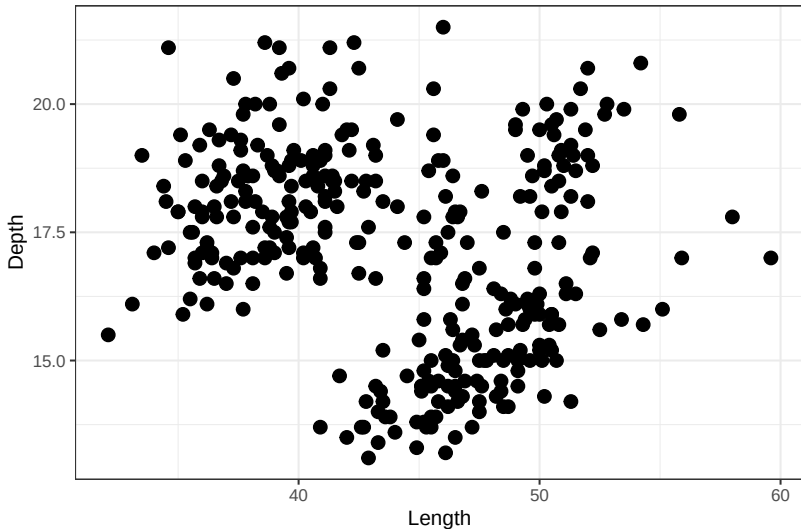
Note: In the raw data, bill dimensions are recorded as "culmen length" and "culmen depth". The culmen is the dorsal ridge atop the bill.

SIMPSON'S PARADOX

```
penguin_df <-  
  palmerpenguins::penguins %>%  
    na.omit()  
head(penguin_df)
```

```
## # A tibble: 6 x 8  
##   species island   bill_length_mm bill_depth_mm flipper_length_mm body_mass_g  
##   <fct>   <fct>         <dbl>         <dbl>         <int>         <int>  
## 1 Adelie  Torgersen         39.1          18.7          181          3750  
## 2 Adelie  Torgersen         39.5          17.4          186          3800  
## 3 Adelie  Torgersen         40.3          18           195          3250  
## 4 Adelie  Torgersen         36.7          19.3          193          3450  
## 5 Adelie  Torgersen         39.3          20.6          190          3650  
## 6 Adelie  Torgersen         38.9          17.8          181          3625  
## # i 2 more variables: sex <fct>, year <int>
```

SIMPSON'S PARADOX



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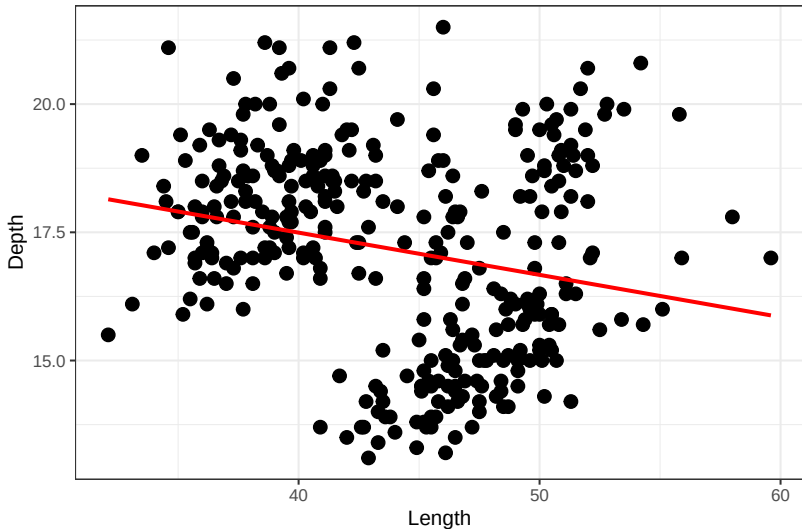


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SIMPSON'S PARADOX



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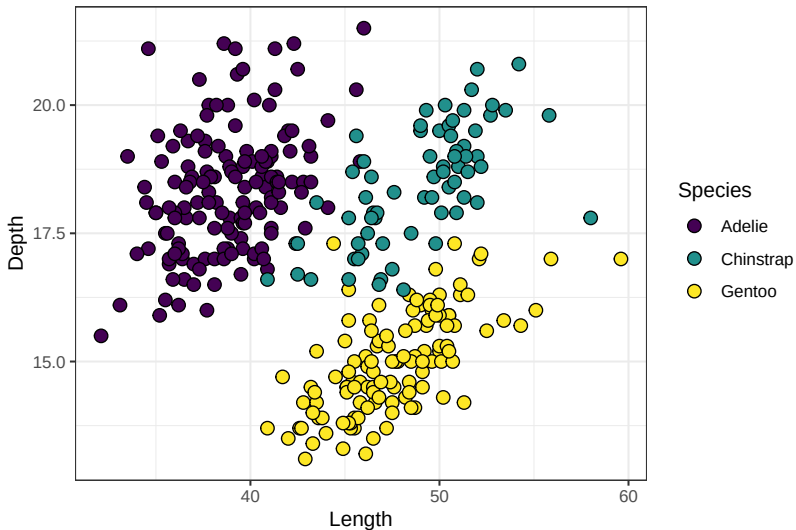


SIMPSON'S PARADOX

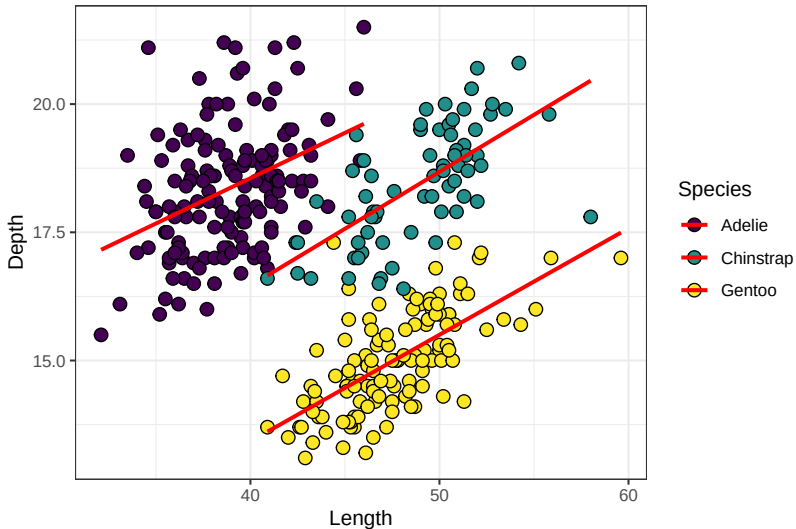
```
summary(lm(bill_depth_mm ~ bill_length_mm, data = penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ bill_length_mm, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.1548 -1.4291  0.0122  1.3994  4.5004
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   20.78665    0.85417   24.335 < 2e-16 ***
## bill_length_mm -0.08233    0.01927   -4.273 2.53e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.92 on 331 degrees of freedom
## Multiple R-squared:  0.05227,    Adjusted R-squared:  0.04941
## F-statistic: 18.26 on 1 and 331 DF,  p-value: 2.528e-05
```

SIMPSON'S PARADOX



SIMPSON'S PARADOX



SIMPSON'S PARADOX

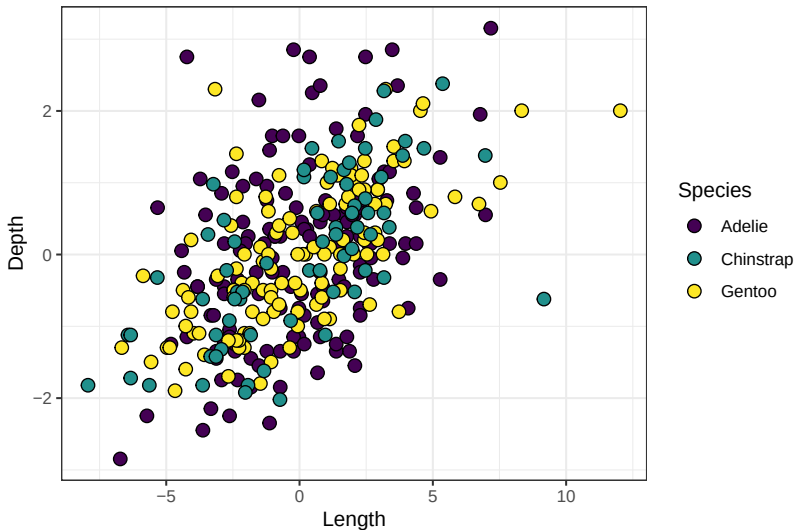
```
summary(lm(bill_depth_mm ~ bill_length_mm + species, data = penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ bill_length_mm + species, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.4579 -0.6814 -0.0431  0.5441  3.5994
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   10.56526     0.69093   15.291 < 2e-16 ***
## bill_length_mm    0.20044     0.01768   11.337 < 2e-16 ***
## speciesChinstrap -1.93308     0.22572  -8.564 4.26e-16 ***
## speciesGentoo   -5.10332     0.19440 -26.252 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9543 on 329 degrees of freedom
## Multiple R-squared:  0.7673, Adjusted R-squared:  0.7652
## F-statistic: 361.6 on 3 and 329 DF,  p-value: < 2.2e-16
```

SIMPSON'S PARADOX

```
penguin_df <- penguin_df %>%  
  group_by(species) %>%  
  mutate(bill_depth_mm_demeaned =  
    bill_depth_mm - mean(bill_depth_mm),  
    bill_length_mm_demeaned =  
    bill_length_mm - mean(bill_length_mm))
```

SIMPSON'S PARADOX



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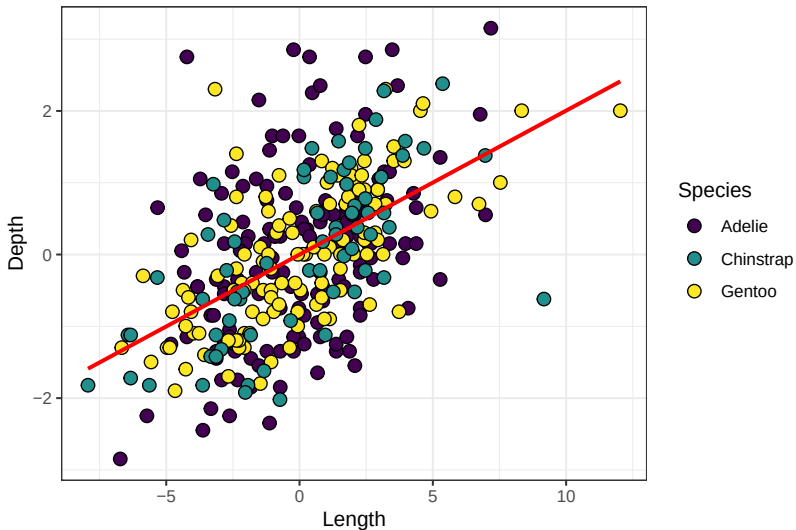


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SIMPSON'S PARADOX



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SIMPSON'S PARADOX

```
summary(lm(bill_depth_mm_demeaned ~ bill_length_mm_demeaned, data = penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm_demeaned ~ bill_length_mm_demeaned,
##     data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.4579 -0.6814 -0.0431  0.5441  3.5994
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.979e-16  5.214e-02   0.00    1
## bill_length_mm_demeaned 2.004e-01  1.763e-02  11.37 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9514 on 331 degrees of freedom
## Multiple R-squared:  0.2809, Adjusted R-squared:  0.2788
## F-statistic: 129.3 on 1 and 331 DF,  p-value: < 2.2e-16
```

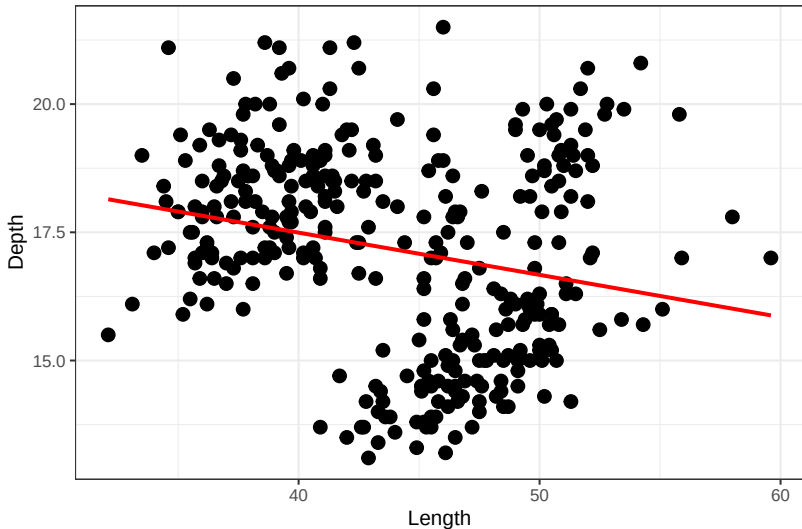
SIMPSON'S PARADOX

Simpson's paradox

- ▶ It's not really a “paradox” - it just depends what *specific comparison* you're interested in and how you are trying to *generalize*!



SIMPSON'S PARADOX



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SIMPSON'S PARADOX

Simpson's paradox

- ▶ It is an empirical fact that across all penguins, bill depth is negatively associated with bill length. This is not itself “wrong” in some empirical sense - it only becomes wrong when we start generalizing that all-species pattern to any specific species.



SIMPSON'S PARADOX

Simpson's paradox

- ▶ It is an empirical fact that across all penguins, bill depth is negatively associated with bill length. This is not itself “wrong” in some empirical sense - it only becomes wrong when we start generalizing that all-species pattern to any specific species.
- ▶ Whether this is a “problem” depends on the research question; in this case, the all-species pattern, while empirically measurable, is likely not *useful* for any research question.

SIMPSON'S PARADOX

- ▶ In the early 1970s, UC Berkeley was sued for gender discrimination over admission to graduate school.



SIMPSON'S PARADOX

	All		Men		Women	
	Applicants	Admitted	Applicants	Admitted	Applicants	Admitted
Total	12,763	41%	8,442	44%	4,321	35%



SIMPSON'S PARADOX

- ▶ Women disproportionately applied to departments with low acceptance rates while men disproportionately applied to departments with high acceptance rates.



SIMPSON'S PARADOX

Department	All		Men		Women	
	Applicants	Admitted	Applicants	Admitted	Applicants	Admitted
A	933	64%	825	62%	108	82%
B	585	63%	560	63%	25	68%
C	918	35%	325	37%	593	34%
D	792	34%	417	33%	375	35%
E	584	25%	191	28%	393	24%
F	714	6%	373	6%	341	7%
Total	4526	39%	2691	45%	1835	30%

Legend:

greater percentage of successful applicants than the other gender

greater number of applicants than the other gender

bold - the two 'most applied for' departments for each gender

SIMPSON'S PARADOX

- ▶ What do we make of this?



SIMPSON'S PARADOX

- ▶ What do we make of this?
- ▶ It depends what we care about and our research question.



SIMPSON'S PARADOX

- ▶ What do we make of this?
- ▶ It depends what we care about and our research question.
- ▶ Gender wage gap example.



REGRESSION WITH MULTIPLE VARIABLES

Causal inference

- ▶ People who drink more wine live longer.